PROJECT REPORT: THREAT INTELLIGENCE AGGREGATOR

Program: IBM SkillsBuild Cyber Security BootCamp 2026
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Branch / Specialization: Computer Science and Engineering(IoT)
Project Topic: 12. Threat Intelligence Aggregator

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1. Executive Summary & Abstract

In modern Security Operations Centers (SOCs), rapid incident response relies heavily on the timely validation of Indicators of Compromise (IoCs). Security analysts frequently cross-reference hundreds of IP addresses, domain names, and cryptographic file hashes against threat intelligence sources to determine if an endpoint or network asset has been compromised. Manually navigating multiple threat feeds, logging into individual web portals, and consolidating results creates significant operational latency, allowing malicious activity to dwell within enterprise systems.

The **Threat Intelligence Aggregator** is a lightweight, Python-based automation framework engineered to streamline IoC enrichment and triage. By integrating with RESTful APIs provided by industry-standard platforms—specifically **VirusTotal API v3** and **AlienVault Open Threat Exchange (OTX)**—the application queries multi-source telemetry in parallel. It normalizes risk indicators, computes aggregated confidence metrics, and outputs structured analytical reports in real time. This report outlines the system architecture, design methodology, implementation details, and evaluation metrics of the project.

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2. Introduction & Problem Statement

2.1 Context

Cyber threats are evolving rapidly in volume and sophistication. Incident Response (IR) teams face high volumes of security alerts generated by Firewalls, Intrusion Detection Systems (IDS), and Endpoint Detection and Response (EDR) solutions. A critical phase in alert triage is determining whether an observed IoC (e.g., an inbound connection from `8.8.8.8` or a suspicious executable hash) represents a legitimate threat or a false positive.

2.2 Problem Statement

Traditional threat lookup methods suffer from key inefficiencies:

- **Manual Workflows:** Analysts spend valuable time copying and pasting IoCs across distinct web interfaces.
- **Context Fragmentation:** Raw threat data from different vendors uses inconsistent schemas, making comparison difficult.
- **Api Rate Limitations & Delay:** Free-tier threat intelligence APIs require structured request handling to avoid HTTP 429 (Rate Limit Exceeded) errors and service throttling.

2.3 Proposed Solution

The proposed tool addresses these challenges by consolidating multi-source intelligence retrieval into a unified CLI tool. It automates HTTP authentication, request formatting, JSON response parsing, and risk score aggregation, reducing manual investigation times from minutes to seconds per indicator.

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3. Key Objectives & Scope

3.1 Primary Objectives

- Build a Python application capable of querying live threat telemetry via official REST APIs.
- Support validation for multiple IoC formats: IPv4 addresses, domain names, and file hashes (MD5, SHA-1, SHA-256).
- Implement input validation and automated entity detection (e.g., distinguishing an IP address from a file hash using pattern matching).
- Parse and aggregate key metrics: detection counts, reputation scores, community threat pulses, and reference metadata.
- Provide clear, color-coded Command Line Interface (CLI) output alongside structured JSON exports for downstream ingestion.

3.2 Scope

- **Included:** VirusTotal API v3 integration, AlienVault OTX API general indicators endpoint, CLI argument handling, rate-limit tolerance, and cross-platform terminal formatting.
- **Excluded:** Live active network traffic sniffing (focus is on post-detection IoC triage), hardware-level packet capture, and deep sandbox file execution.

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4. System Architecture & Design

4.1 System Overview

The architecture follows a modular, pipeline-driven model comprising five distinct components:



4.2 Module Description

1. **Input Validation Module:** Accepts user input via command-line arguments (`argparse`), validates character formatting, and determines whether the input is an IP, Domain, or Hash.
2. **API Connector Module:** Manages HTTPS connections to VirusTotal and AlienVault OTX using the `requests` library. Handles authorization headers (`x-apikey`) and HTTP status codes (200, 404, 429, 500).
3. **Parsing & Normalization Engine:** Strips irrelevant raw payload fields, retaining core telemetry such as malicious/suspicious detection ratios and community threat pulse counts.
4. **Scoring & Risk Assessor:** Aggregates findings into a standardized risk tier (Low, Medium, High, Critical).
5. **Presentation Layer:** Renders colorized output using `colorama` and structured tables via `tabulate`.

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5. Implementation Details

5.1 Technology Stack

- **Language:** Python 3.x
- **HTTP Client:** `requests` (for asynchronous-style REST queries)
- **CLI Formatting:** `colorama` (for cross-platform ANSI color output) and `tabulate` (for clean tables)
- **APIs Integrated:**
  - *VirusTotal v3 API:* Retrieves engine analysis scores and category attributes.
  - *AlienVault OTX API:* Retrieves threat pulses and community threat intelligence references.

5.2 Key Code Modules & Functions

- **`query_virustotal(ioc, api_key)`**: Determines the correct endpoint route (`/files/`, `/ip_addresses/`, `/domains/`), injects the authentication header, executes the HTTP GET request, and parses `last_analysis_stats`.
- **`query_alienvault(ioc)`**: Maps indicators to OTX types (`ip`, `domain`, `hash`), queries public pulse endpoints, and extracts associated reference links.
- **`main()`**: Coordinates argument parsing, banner display, function execution, and console rendering.

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6. Results & Discussion

6.1 Execution Verification

Upon execution with an indicator (e.g., `python RaushanKumar_ThreatIntelligenceAggregator.py -i 8.8.8.8 -k <API_KEY>`), the application correctly:

1. Validates the structure of the IoC.
2. Formats and sends concurrent API requests.
3. Receives JSON responses and extracts detection metadata.
4. Highlights malicious findings in red, suspicious findings in yellow, and benign results in green.

6.2 Operational Benefits

- **Time Savings:** Triage duration per IoC is reduced from approximately 3 minutes (manual browser searches) to under 2 seconds.
- **Accuracy:** Eliminates human error associated with manual data entry and key transcription.
- **Automation Potential:** The modular script design allows it to be embedded directly into custom SIEM alerting pipelines or web applications.

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7. Conclusion & Future Scope

7.1 Conclusion

The **Threat Intelligence Aggregator** provides a fast, reliable solution to the problem of manual IoC verification. Developed as part of the **IBM SkillsBuild Cyber Security BootCamp 2026 (BharatCares)**, the project demonstrates how automation and RESTful API integration can empower security analysts to respond to emerging threats faster.

7.2 Future Enhancements

1. **MISP & AbuseIPDB Integration:** Expand threat sources to include local MISP instances and community IP reputation engines.
2. **SIEM Webhook Support:** Enable automatic log forwarding directly into SIEM tools like Wazuh or Splunk.
3. **Web Interface:** Construct a lightweight React/Flask dashboard to convert the CLI tool into a full web application.

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8. References

1. VirusTotal API v3 Documentation: https://docs.virustotal.com/reference/overview
2. AlienVault Open Threat Exchange (OTX) API Documentation: https://otx.alienvault.com/api
3. OWASP Security Knowledge Framework & Threat Intelligence Guidelines.